

Engineering Design & Problem Solving (InSPIRESS)

TEKS §127.785 • Advanced • Senior • Mr. Gavaldon • Da Vinci School

Week 2
MONDAY Day 6 - Launch Fusion + First Sketch

Aug 17

OBJECTIVE

Students will navigate the Fusion 360 interface and create a dimensioned sketch.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - industry-standard CAD (this is the shared Semester-1 Fusion foundation)

ELPS / EB SUPPORT

Steps mirrored on-screen + in Classroom screenshots (VC). Finishers help peers (PN).

AGENDA • 45 MIN

5' Bell-work: what is CAD?

12' Fusion interface + watch intro

23' Guided: your first sketch

5' Save + exit

SLIDES / ACTIVITY

Into Fusion 360

NOTEBOOK

Sketch the Fusion screen and label the toolbar, sketch button, and timeline.

TURN IN

— (notebook / no turn-in)

TUESDAY Day 7 - Sketch to Part

Aug 18

OBJECTIVE

Students will extrude a sketch into a solid and apply basic modifications.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - CAD - solid modeling (shared Fusion foundation)

ELPS / EB SUPPORT

Live demo before work time (VC, RS); one-page step card at each station (CD).

AGENDA • 45 MIN

5' Bell-work: what does 'extrude' mean?

12' Extrude, fillet, modify

23' Turn your sketch into a solid

5' Save + exit

SLIDES / ACTIVITY

Make It Solid

NOTEBOOK

List the 3 steps you used to turn a sketch into a solid.

TURN IN

— (notebook / no turn-in)

WEDNESDAY Day 8 - Model a Simple Object

Aug 19

OBJECTIVE

Students will plan and model a simple real object to scale in Fusion 360.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - CAD - model a real object to dimension (shared Fusion foundation)

ELPS / EB SUPPORT

Plan-before-model sheet lowers language load (CD). Pair modeling for support (PN).

AGENDA • 45 MIN

5' Bell-work: pick a simple object to model

10' Plan it: shapes + dimensions

25' Model your object in Fusion

5' Save + exit

SLIDES / ACTIVITY

Model Something Real

NOTEBOOK

Sketch your object with real dimensions before you model it.

TURN IN

Model Planning Sheet

THURSDAY

Day 9 - Design Choices + Prep to Present

Aug 20

OBJECTIVE

Students will prepare a one-minute presentation explaining their object and its design choices.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - technical communication - presenting a design & its rationale

ELPS / EB SUPPORT

Pitch planner with sentence frames (CD, RS). Rehearse in pairs before the real thing (PN, WT).

AGENDA • 45 MIN

5' Bell-work: why did you design it that way?

12' How to talk about design choices

20' Build your 1-minute pitch

8' Rehearse with your team

SLIDES / ACTIVITY

Talk About Your Design

NOTEBOOK

Write your 1-minute pitch outline: what it is, why these choices, one thing you'd improve.

TURN IN

1-Minute Pitch Planner

FRIDAY

Day 10 - First Team Presentations

Aug 21

OBJECTIVE

Students will present their Fusion object and design choices and give constructive peer feedback.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - deliver a technical presentation; give & receive feedback

ELPS / EB SUPPORT

Feedback uses a simple 'glow + grow' frame (CD). Presenting your own work with a rehearsed script supports EB students (RS, WT).

AGENDA • 45 MIN

5' Bell-work: one thing a good listener does

8' How we give feedback

27' Present your object (1 min each)

5' Feedback + exit

SLIDES / ACTIVITY

Present It

NOTEBOOK

Log your pitch (what went well, one thing to improve) + one piece of feedback you got.

TURN IN

Presentation Peer-Feedback